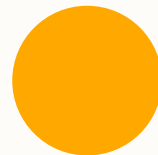




Knowledge and Expert System

Reese Hatfield



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Knowledge

- We've talked roughly about how different generations of Artificial intelligence work
- Different approaches throughout the years





Knowledge

- Most of these systems are still in place
- Domain-specific problem
- Deep learning is good at specific things
- Same for rule based systems





Knowledge

- Hard to go into the nitty-gritty in a course like this
- Especially for newer generations
- Older systems are simpler
- Rule Based systems
- Let's engineer this from scratch





Knowledge

- We want our systems to be smart
- It needs to know things
- We should be able to:
 - Ask a questions
 - It should consult a bunch of knowledge
 - Reply with some answer





Knowledge

- Remember computers run on plaintext
- We need some way of encoding knowledge first
 - There's not one solution
- But we need to agree how knowledge works first





Knowledge

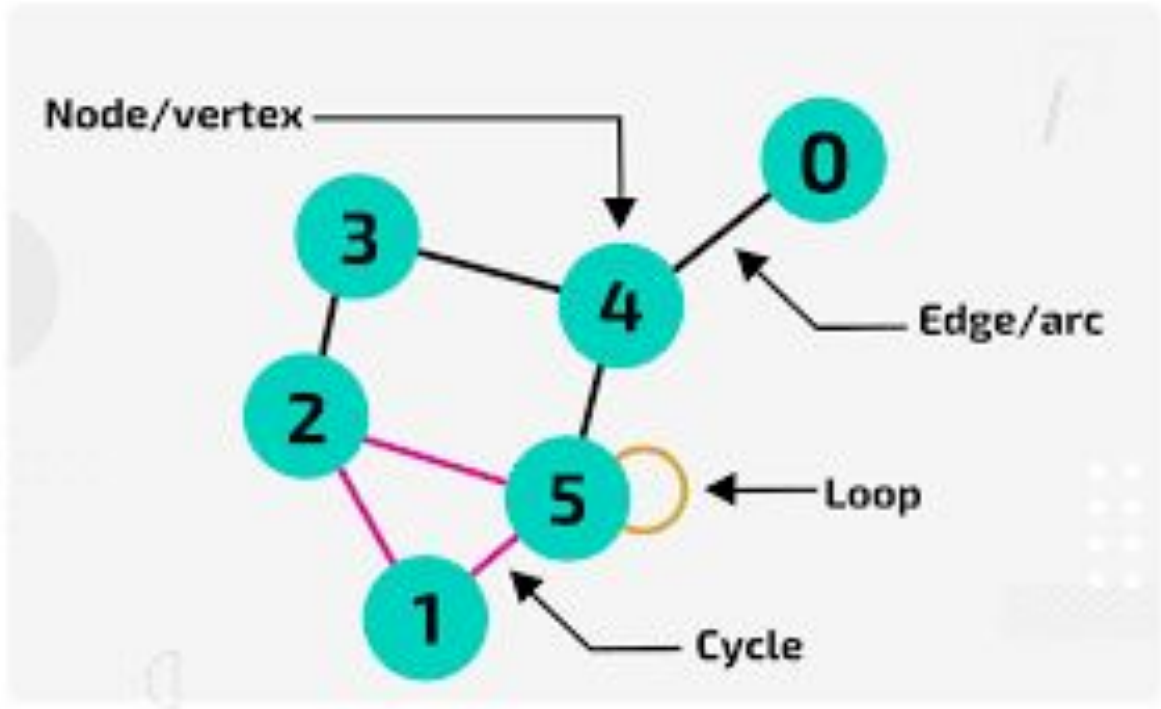
- Aside: Graphs
- Not charts
- A graph in computer science:
 - A series of nodes, connected with edges
- This is the core of most things in computer science





Knowledge

- A lot of terms:
- **Node**
- **Edge**
- Are what all need





Knowledge

- We would talk more about why this is important if we did more programming
- Networks are graphs
- Lists are graphs
- Files are graphs





Knowledge

- Network as a graph:
- Node $\Rightarrow$ Any device
- Edge $\Rightarrow$ Any connection between those devices
- Graph is the general structure





Knowledge

- Nodes often have names
 - Sometimes not
- Edges often have something else attached to them
 - Weights
 - Names
 - Direction?





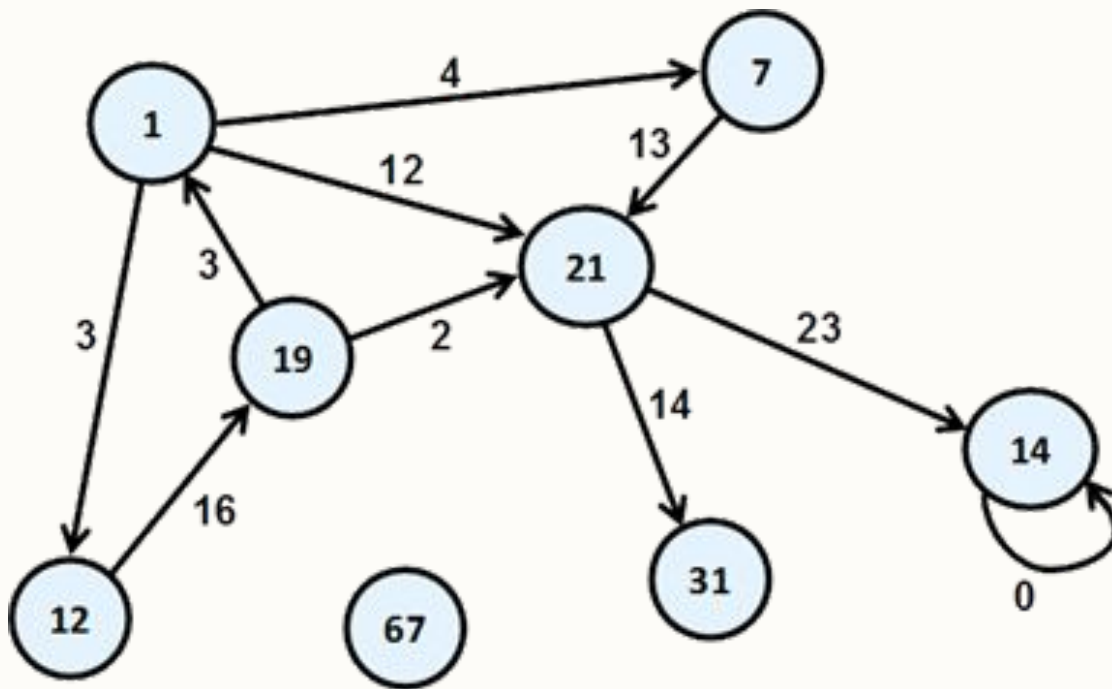
Knowledge

- Graph edges are also sometimes **directed**
- Meaning they only go one direction
- Can imply transience if specified
- Sometimes they can go both directions if specified





Knowledge





Knowledge

- Generally speaking:
 - Weights are more common to see
- Names are often termed "labels"
- A label would let us directly indicate a relationship





Knowledge

Charlie

Derek

Joey



Knowledge

Charlie

Derek

Joey

Alice

?

Bob



Knowledge

Charlie

Derek

Joey

Alice

Siblings

Bob



Knowledge

Charlie

Derek

Joey

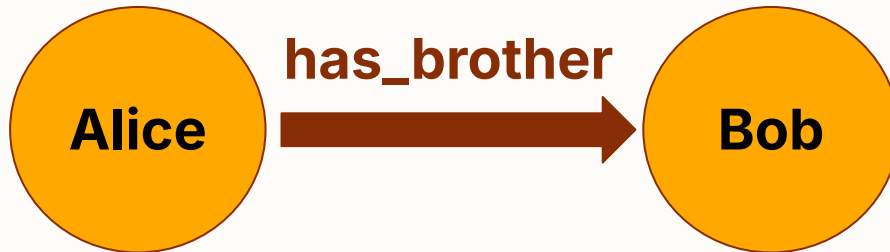
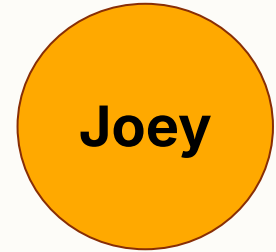
Alice

has_brother

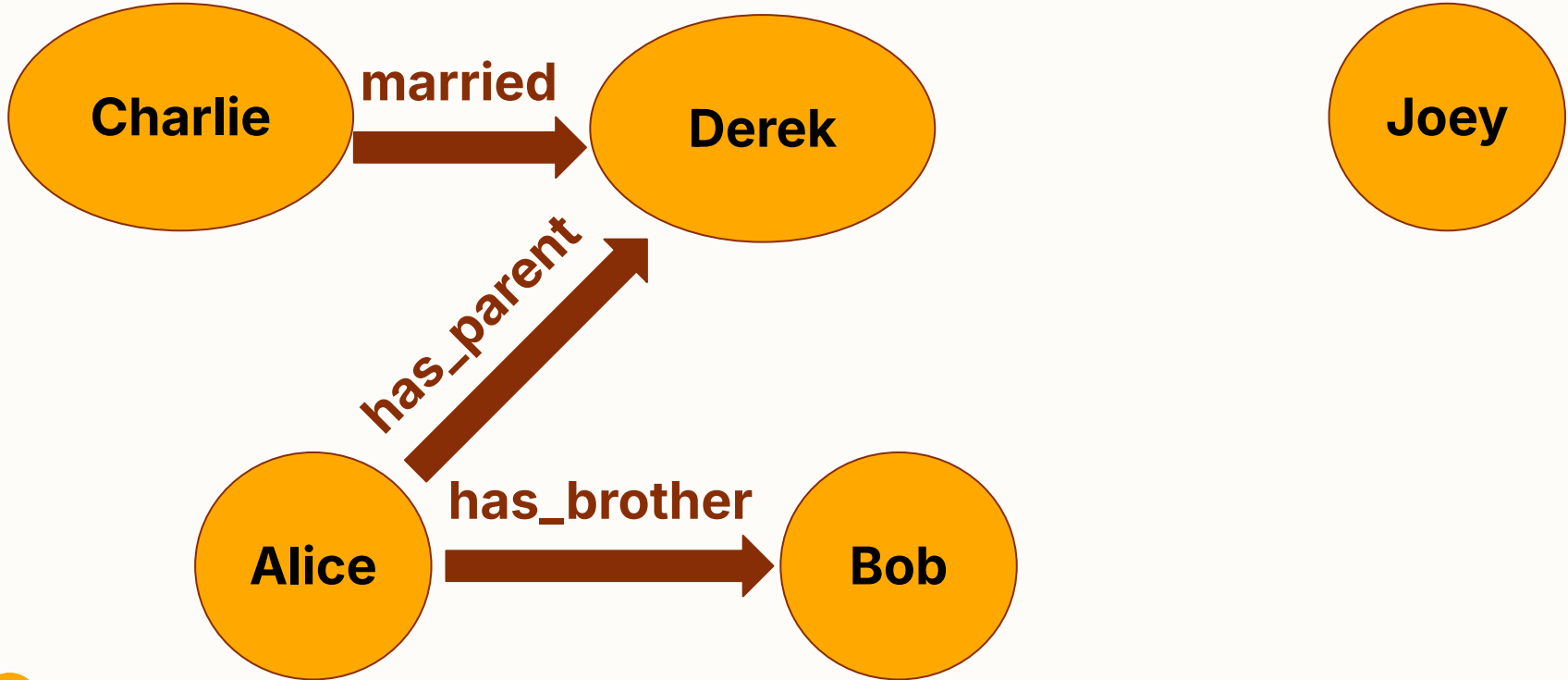
Bob



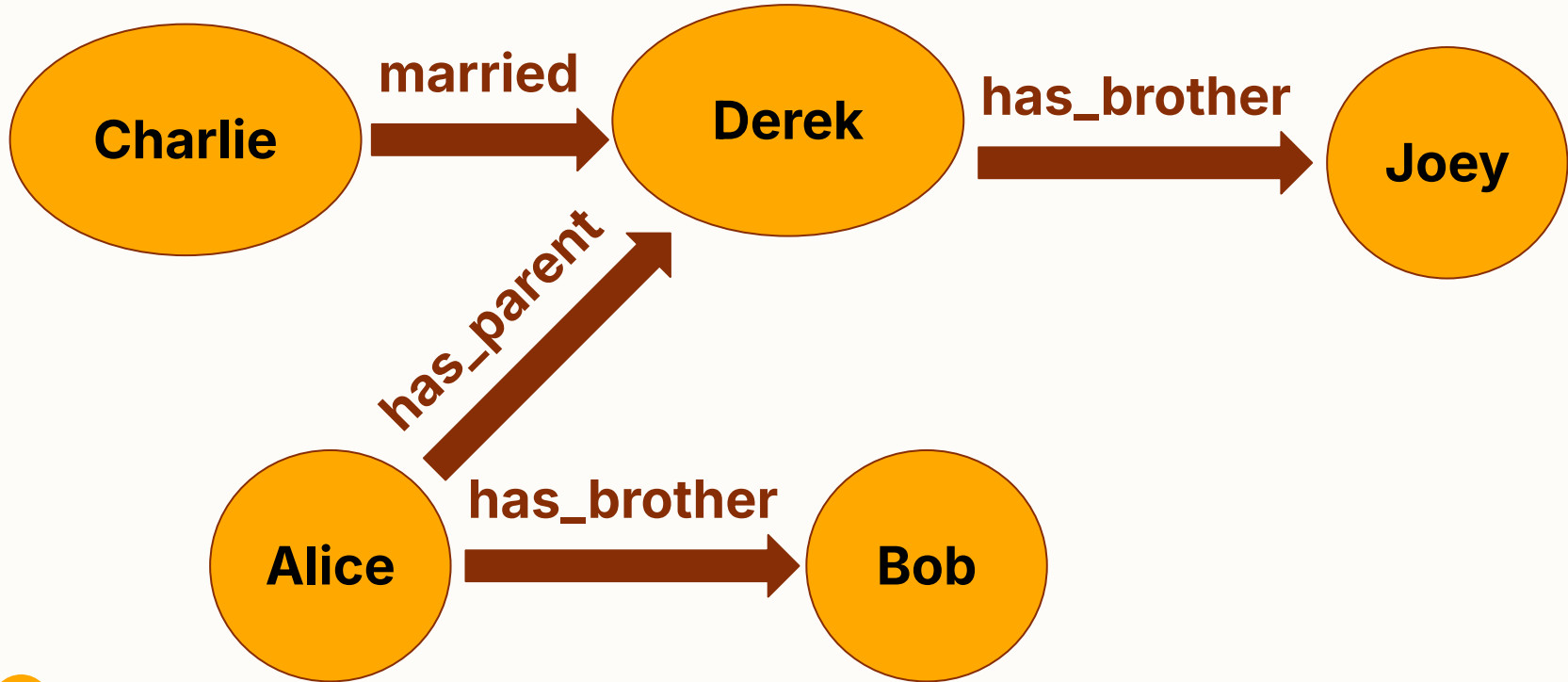
Knowledge



Knowledge



Knowledge





Knowledge

- Labels also have specific relationships
 - Often bridged with deep learning
- I could add more
- Some of these are multi-directional
- has_sister
- etc.





Knowledge

- Build a series of facts
- **Subject, Relationship, Object**
- Alice, has_brother, Bob
- Encode them in some way
- We can act upon programmatically, e.g. Javascript





Knowledge

- Query the data in some way
- In the same way you “query” google or GPT or the like
- Who is Alice's brother?
- Consult the graph
- Usually encoded in a different manner





Knowledge

- Write code that would programmatically
- Iterate thru the data
- Extract a **Factual** relationship

Alice, has_brother, Bob
Alice, has_parent, Derek
Charlie, married, Derek
Derek, has_brother, Joey





Knowledge

- Could also make inferences!
- Who is Bob's parent?
- Label relationships
- If we have the knowledge encoded

Alice, has_brother, Bob
Alice, has_parent, Derek
Charlie, married, Derek
Derek, has_brother, Joey





Knowledge

- Often called **Semantic Networks**
- Used to build rule-based expert systems
- Specifically useful to pull out **FACTUAL** results
- Can see how it derived a conclusion





Knowledge

- Compare this to something like GPT
- Predicting a next token
- But its not actually sourcing that information
- Unless it uses RAG (still iffy?!)





Knowledge

- Rule-based expert systems were the standard for many years
- Today, they are mostly used in conjunction with deep learning
- (Often, on TOP of)





Knowledge

- Difficult to demonstrate an easily queryable semantic network
 - Something interactive
- Since it's often used on TOP of, we can just use our plaintext encoding in an LLM prompt
- Sacrifices factuality, for demonstration





Knowledge

- KASTLE Lab
- The Knowledge and Semantic Technologies
- Researches this exact topic
- Leading Research lab for this kind of thing
- Cogan Shimizu

